

SC2000 Probability & Statistics for Computing  
NTU BSc Computer Science — Comprehensive Textbook (0–100)

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# Chapter 1

## Counting & Combinatorics

*“Counting is the oldest branch of computing.”* Before probability, before statistics, before machine learning: how many ways can it happen? Every algorithm that enumerates, every cryptographic key space, every dataset split begins with counting.

**From zero: no prior combinatorics assumed.** We build from “how many?” with pictures before formulas. If you can count on your fingers, you can follow this chapter. Every rule is motivated by a concrete computing scenario.

### Learning Outcomes

After this chapter you will be able to:

- (L1) apply the multiplication (product) and addition (sum) rules to count compound choices;
- (L2) count permutations and combinations with and without repetition;
- (L3) use stars-and-bars, the binomial theorem, and inclusion–exclusion;
- (L4) apply the pigeonhole principle to prove existence claims;
- (L5) estimate key-space sizes and algorithmic branching factors.

### 1.1 Why Counting Matters in Computing

How many 8-character passwords over 62 symbols ( $a$ – $z$ ,  $A$ – $Z$ ,  $0$ – $9$ )? How many ways to choose 5 features from 50? How many binary strings of length  $n$  have no two consecutive 1s? Each is a counting problem whose answer determines security, complexity, or feasibility. We build tools systematically.

### 1.1.1 Intuition: Choices in Sequence

Imagine ordering lunch:  $3 \text{ mains} \times 4 \text{ drinks} = 12 \text{ combos}$ . Draw a tree: 3 branches, each splits into 4. Counting leaves = counting outcomes. This picture *is* the product rule.

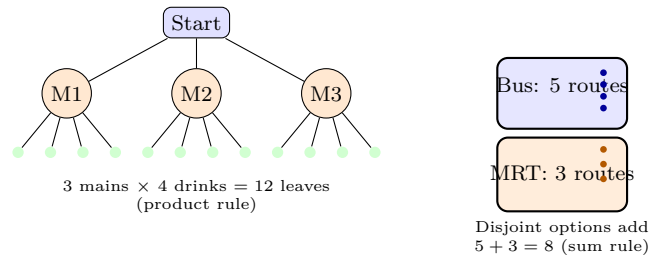


Figure 1.1: Left: product rule as a branching tree. Right: sum rule as disjoint piles.

## 1.2 Product and Sum Rules

**Theorem 1.1** (Product rule). If a task has  $k$  sequential stages with  $n_1, \dots, n_k$  choices respectively (choices at stage  $i$  may depend on earlier choices but count is  $n_i$ ), total outcomes =  $n_1 n_2 \cdots n_k$ .

**Theorem 1.2** (Sum rule). If sets  $A_1, \dots, A_k$  are pairwise disjoint,  $|A_1 \cup \cdots \cup A_k| = |A_1| + \cdots + |A_k|$ . For overlapping sets, see inclusion–exclusion (§1.5).

**Example 1.1** (Passwords). An 8-character password over 62 symbols:  $62^8 \approx 2.18 \times 10^{14}$  possibilities. At  $10^9$  guesses/sec,  $\approx 2.5$  days to brute-force. Product rule gives key-space size directly.

### 1.2.1 Counting with Cases

Often we partition by a property, count each case, then sum. This is sum rule in action. E.g., binary strings length  $n$  with exactly  $k$  ones: choose positions for the  $k$  ones (see §1.4).

## 1.3 Permutations

A *permutation* is an ordered arrangement.

**Definition 1.1** (Permutation). The number of ordered arrangements of  $k$  distinct items chosen from  $n$  distinct items is

$$P(n, k) = \frac{n!}{(n-k)!} = n(n-1) \cdots (n-k+1).$$

When  $k = n$ ,  $P(n, n) = n!$ .

*Why divide by  $(n-k)!$ ?* Line up all  $n!$  orders; the tail  $n-k$  items beyond position  $k$  are irrelevant—divide out their  $(n-k)!$  orders.

**Example 1.2.** Ranking top 3 out of 10 teams:  $P(10, 3) = 10 \cdot 9 \cdot 8 = 720$ . PIN codes of 4 distinct digits from 10:  $P(10, 4) = 5040$ .

**Permutations with repetition.** If  $n_1$  identical of type 1,  $n_2$  of type 2,  $\dots$ ,  $n_r$  of type  $r$  with  $\sum n_i = n$ ,

$$\frac{n!}{n_1! n_2! \cdots n_r!}.$$

E.g., arrangements of BANANA:  $6!/(3! 2! 1!) = 60$ .

## 1.4 Combinations

**Definition 1.2** (Combination). The number of ways to choose an unordered set of  $k$  from  $n$  distinct items is

$$\binom{n}{k} = \frac{n!}{k! (n-k)!}, \quad 0 \leq k \leq n.$$

Read “ $n$  choose  $k$ ”. By convention  $\binom{n}{0} = 1$ .

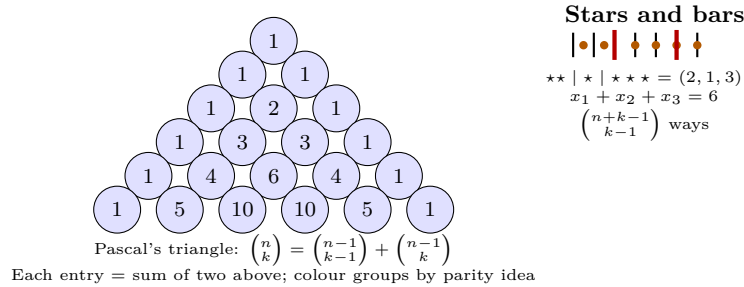


Figure 1.2: Left: Pascal's triangle (binomial coefficients). Right: stars-and-bars encoding.

Key identities:  $\binom{n}{k} = \binom{n}{n-k}$ ,  $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$  (Pascal),  $\sum_{k=0}^n \binom{n}{k} = 2^n$ ,  $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$ .

**Stars and bars.** Nonnegative integer solutions to  $x_1 + \cdots + x_k = n$  correspond to placing  $k-1$  bars among  $n+k-1$  slots:

$$\binom{n+k-1}{k-1}.$$

For positive solutions ( $x_i \geq 1$ ), shift  $y_i = x_i - 1$ :  $\binom{n-1}{k-1}$ . Example: distributing 10 identical tasks to 4 servers:  $\binom{13}{3} = 286$  ways.

## 1.5 Binomial Theorem and Inclusion–Exclusion

**Theorem 1.3** (Binomial theorem). For any  $x, y$  and  $n \geq 0$ ,

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Setting  $x = y = 1$  gives  $\sum_k \binom{n}{k} = 2^n$ . Setting  $x = 1, y = -1$  gives alternating sum 0.

**Example 1.3** (Committee with a chair). Choose a  $k$ -person committee from  $n$  and a chair among them:

$k \binom{n}{k} = n \binom{n-1}{k-1}$  ways (pick chair first vs. committee first). Summing over  $k$  gives  $\sum_k k \binom{n}{k} = n2^{n-1}$ , a useful identity for expected value of  $\text{Bin}(n, 1/2)$ .

**Theorem 1.4** (Inclusion–exclusion, two and three sets).

$$|A \cup B| = |A| + |B| - |A \cap B|, \quad |A \cup B \cup C| = \sum |A_i| - \sum |A_i \cap A_j| + |A \cap B \cap C|.$$

In general, add singles, subtract pairs, add triples, . . . .

**Example 1.4** (Derangements via inclusion–exclusion). Counting surjections or derangements uses the full alternating sum; e.g., onto functions  $[n] \rightarrow [k]$  count  $= k! S(n, k) = \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n$ .

## 1.6 The Pigeonhole Principle

**Theorem 1.5** (Pigeonhole principle). If  $n$  pigeons go into  $k$  holes with  $n > k$ , some hole has  $\geq 2$  pigeons. Generalised: some hole has  $\geq \lceil n/k \rceil$  pigeons.

This proves *existence* without construction. In computing: with  $2^n + 1$  files each  $n$  bits, two share the same  $n$ -bit hash (collision). With 367 people, two share a birthday (including Feb 29). In any 6 people, 3 mutually know each other or 3 mutually strangers (Ramsey  $R(3, 3) = 6$ ).

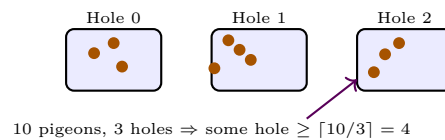


Figure 1.3: Pigeonhole principle: more pigeons than holes forces a crowded hole.

## 1.7 Counting in Code

Counting formulas become code when enumerating. Python's `math.comb` and `math.perm` implement  $\binom{n}{k}$  and  $P(n, k)$  exactly with big integers.

```

1  import math
2  # key-space and combinations in computing
3  print(math.perm(10, 3))           # 720 rankings
4  print(math.comb(50, 5))           # 2_118_760 feature subsets
5  print(math.comb(10+4-1, 4-1))     # 286 task distributions (stars/bars)
6  # brute-force estimate: 62^8 passwords
7  print(62**8, " ~ 2.2e14")
8  # verify binomial identity sum_k C(n,k) == 2^n
9  n = 20
10 assert sum(math.comb(n, k) for k in range(n+1)) == 2**n

```



## 1.8 Chapter Notes

Dirichlet stated the pigeonhole (Schubfach) principle in 1834. Stars-and-bars is folk combinatorics; see Feller Vol. 1. Binomial coefficients appear in Knuth TAOCP Vol. 1 and CLRS App. C. For large  $n$ , Stirling  $n! \sim \sqrt{2\pi n}(n/e)^n$  estimates key spaces without overflow.

## 1.9 Exercises

- E1.1 **Product vs. sum.** A system offers 4 operating systems, each with 6 desktop environments, plus 3 standalone window managers usable with any OS. How many distinct software stacks? Identify where product vs. sum applies and compute the total.
- E1.2 **Permutations with repetition.** How many distinct strings can be formed from MISSISSIPPI? Generalise to a multiset with counts  $n_1, \dots, n_r$  and verify your formula gives 60 for BANANA.
- E1.3 **Stars and bars.** (a) How many nonnegative integer solutions to  $x_1 + x_2 + x_3 + x_4 = 12$ ? (b) How many *positive* solutions? (c) A load balancer distributes 12 identical requests to 4 servers, each must get  $\geq 1$ . Count assignments.
- E1.4 **Inclusion–exclusion.** Among 100 students, 60 know Python, 50 know Java, 30 know both. How many know neither? Extend: with a third language (40 know C, pairwise overlaps 20, 15, 18, triple 8), how many know at least one?
- E1.5 **Pigeonhole in hashing.** A hash table has 1000 buckets and stores 1001 distinct keys with a deterministic hash function. Prove a collision must occur. If 5000 keys are stored, give a lower bound on the maximum bucket occupancy.

## Chapter 2

# Sample Spaces & Events

*“Probability is the logic of uncertainty.”* We cannot predict a single coin toss, but we can describe perfectly the *space* of all possible tosses and assign consistent weights to events. That space and those weights are the foundation of everything that follows.

**From zero: sets  $\rightarrow$  sample spaces.** We assume only sets and counting (Ch. 1). Probability is built as *counting with weights*: favourable over possible, generalised to non-uniform weights via axioms.

### Learning Outcomes

After this chapter you will be able to:

- (L1) formalise random experiments via sample spaces, events, and set operations;
- (L2) apply Kolmogorov’s axioms and derive basic probability rules;
- (L3) compute probabilities in equally likely (uniform) models via counting;
- (L4) use complements, unions, and inclusion–exclusion for events;
- (L5) distinguish disjoint vs. independent events and avoid the confusion.

## 2.1 Random Experiments and Sample Spaces

A *random experiment* is any procedure with uncertain outcome but a well-defined set of possibilities.

**Definition 2.1** (Sample space, event). The *sample space*  $\Omega$  is the set of all possible outcomes. A point  $\omega \in \Omega$  is an *outcome*. An *event*  $A \subseteq \Omega$  is a set of outcomes. The *certain event* is  $\Omega$ , the *impossible event* is  $\emptyset$ .

Examples: coin toss  $\Omega = \{H, T\}$ ; two tosses  $\Omega = \{HH, HT, TH, TT\}$ ; die  $\Omega = \{1, \dots, 6\}$ ; packet delay  $\Omega = [0, \infty)$  (continuous); infinite coin tosses  $\Omega = \{H, T\}^\infty$ .

**Events as sets.** “At least one head in two tosses” corresponds to  $\{HH, HT, TH\}$ . “Sum of two dice = 7” corresponds to  $\{(1, 6), (2, 5), \dots, (6, 1)\}$  (6 outcomes out of 36). Set operations:  $A \cap B$  (both),  $A \cup B$  (at least one),  $A^c = \Omega \setminus A$  (not  $A$ ),  $A \setminus B$  ( $A$  but not  $B$ ).

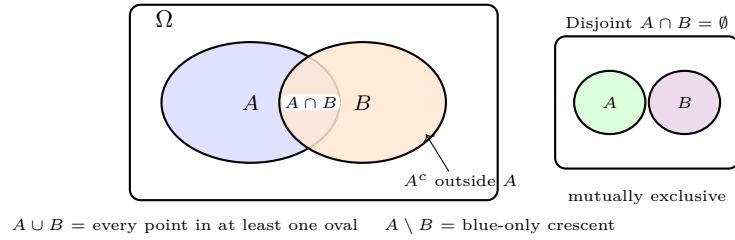


Figure 2.1: Sample space  $\Omega$  as a rectangle; events  $A, B$  as ovals. Set operations become geometry. Right: disjoint (mutually exclusive) events do not overlap.

## 2.2 Kolmogorov’s Axioms

We assign each event a number  $P(A) \in [0, 1]$  obeying:

**Definition 2.2** (Probability axioms). A function  $P$  on events is a *probability measure* if

- (A1)  $P(A) \geq 0$  for every event  $A$ ;
- (A2)  $P(\Omega) = 1$ ;
- (A3) *Countable additivity*: if  $A_1, A_2, \dots$  are pairwise disjoint,  $P(\bigcup_i A_i) = \sum_i P(A_i)$ .

Consequences (prove each from axioms):

- $P(\emptyset) = 0$ ,  $P(A^c) = 1 - P(A)$ .
- If  $A \subseteq B$  then  $P(A) \leq P(B)$  (monotonicity).
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$  (inclusion–exclusion for two events).
- $P(A \cup B) \leq P(A) + P(B)$  (union bound / Boole’s inequality), with equality iff disjoint.
- Continuity: if  $A_n \uparrow A$  then  $P(A_n) \rightarrow P(A)$ .

**Example 2.1** (Why axioms?). Without (A3), we could assign  $P(\{H\}) = 0.6$ ,  $P(\{T\}) = 0.6$  for a coin and get  $P(\Omega) = 1.2 \neq 1$ . Axioms force consistency. They say nothing about *which* numbers—that is modelling.

## 2.3 Equally Likely Outcomes (Uniform Model)

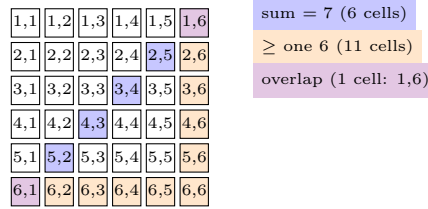
When  $\Omega$  is finite and every outcome is equally likely,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\# \text{ favourable}}{\# \text{ possible}}.$$

This reduces probability to *counting* (Ch. 1). Verify uniformity is justified (fair coin, well-shuffled deck) before using it!

**Example 2.2** (Two dice).  $\Omega$  has 36 ordered pairs.  $A = \{\text{sum } 7\}$  has 6 outcomes, so  $P(A) = 6/36 = 1/6$ .  $B = \{\text{at least one } 6\}$  has 11 outcomes ( $6 + 6 - 1$ ),  $P(B) = 11/36$ .

**Example 2.3** (Birthday problem preview). With  $n$  people and 365 days (ignore Feb 29),  $P(\text{all distinct}) = 365 \cdot 364 \cdots (365 - n + 1)/365^n$ . For  $n = 23$ ,  $P(\text{collision}) \approx 50.7\%$ .



Two dice: 36 ordered pairs. Blue diagonal = 7; orange = at least one 6.

Figure 2.2: Uniform model for two dice: counting favourable cells in the  $6 \times 6$  grid.

## 2.4 Non-Uniform Models and Continuous Spaces

Not every outcome is equally likely. A biased coin:  $P(H) = p$ ,  $P(T) = 1 - p$ . A loaded die: arbitrary  $p_1, \dots, p_6$  summing to 1. Axioms still hold; we just assign weights  $p(\omega)$  with  $\sum_{\omega} p(\omega) = 1$  and  $P(A) = \sum_{\omega \in A} p(\omega)$ .

For continuous  $\Omega$  (e.g.,  $[0, 1]$  for a random number), single points have probability 0; we assign probability to *intervals* via a density. Formally  $P([a, b]) = \int_a^b f(x) dx$  where  $f \geq 0$ ,  $\int_{\Omega} f = 1$ . This previews Ch. 4 (random variables).

## 2.5 Disjoint vs. Independent — A Critical Distinction

**Definition 2.3** (Disjoint).  $A$  and  $B$  are *disjoint* (mutually exclusive) if  $A \cap B = \emptyset$ .

Disjoint is a *set* property (no overlap); it forces  $P(A \cup B) = P(A) + P(B)$ . Independence (Ch. 3) is a *probabilistic* property  $P(A \cap B) = P(A)P(B)$  and depends on  $P$ . Disjoint events with positive probability are *never* independent (if  $A \cap B = \emptyset$  then  $P(A \cap B) = 0 \neq P(A)P(B) > 0$ ). Do not confuse them!

**Example 2.4.** Die roll:  $A = \{1, 2\}$ ,  $B = \{3, 4\}$  are disjoint.  $C = \{1, 3, 5\}$  (odd) and  $D = \{1, 2, 3\}$  are not disjoint (1, 3 overlap) and not independent either ( $P(C \cap D) = 2/6 \neq (3/6)(3/6)$ ).

## 2.6 Why Countable Additivity Matters (countable $\Omega$ )

Finite additivity  $P(A \cup B) = P(A) + P(B)$  for disjoint  $A, B$  is not enough for infinite  $\Omega$  (e.g.,  $\Omega = \mathbb{N}$ ). Countable additivity (A3) lets us sum over countably many disjoint events—needed for geometric distributions  $P(X = k) = (1 - p)^{k-1}p$  where  $\sum_{k \geq 1} P(X = k) = 1$  is an infinite sum. Without it, paradoxes arise (finitely additive “probabilities” that assign 0 to every singleton yet 1 to  $\mathbb{N}$ ). For this book, remember: every infinite disjoint union’s probability is the sum of the infinite series.

## 2.7 Computing Probabilities in Code

```

1  import math, itertools
2
3  # Uniform model: two dice
4  Omega = list(itertools.product(range(1,7), repeat=2))
5  def prob(event): return len(event) / len(Omega)
6
7  sum7 = [x for x in Omega if sum(x)==7]
8  at_least_one6 = [x for x in Omega if 6 in x]
9  print(f"P(sum 7) = {prob(sum7):.4f}")           # 0.1667
10 print(f"P(>= one 6) = {prob(at_least_one6):.4f}") # 0.3056
11 # inclusion-exclusion check
12 both = [x for x in sum7 if 6 in x]
13 print(len(sum7)+len(at_least_one6)-len(both), len(set(sum7)|set(at_least_one6)))
14
15 # Birthday collision for n=23
16 n, days = 23, 365
17 p_distinct = math.perm(days, n) / days**n
18 print(f"P(collision, n=23) = {1-p_distinct:.4f}") # ~0.507

```

## 2.8 Chapter Notes

Kolmogorov axiomatised probability in 1933 (*Grundbegriffe*). The rectangle-and-ovals picture is Venn (1880); the distinction disjoint/independent trips up even experienced practitioners. For continuous spaces the axioms extend via measure theory (see Billingsley); computing needs only the intuition: probability as area under a density.

## 2.9 Exercises

E2.1 **Axioms drill.** From (A1)–(A3) prove:

- (a)  $P(A^c) = 1 - P(A)$ , (b) if  $A \subseteq B$  then  $P(A) \leq P(B)$ ,
- (c)  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .

Which step uses countable additivity?

E2.2 **Uniform counting.** A 5-card hand is dealt from 52 cards. Compute  $P$ (at least one ace) exactly (as a fraction) via the complement. Then compute  $P$ (two pairs) and explain the counting.

E2.3 **Non-uniform.** A biased coin has  $P(H) = 0.6$ . Two independent tosses (independence defined in Ch. 3, assume product rule here). List  $\Omega$  with weights  $p(\omega)$  and compute  $P$ (at least one  $H$ ). Why is  $P(HH) \neq 1/4$ ?

E2.4 **Disjoint vs. independent.** Give an example of (a) disjoint but not independent events, (b) independent but not disjoint events, (c) events that are neither. Prove disjoint + positive probability  $\Rightarrow$  not independent.

E2.5 **Union bound.** Prove Boole's inequality  $P(\bigcup_{i=1}^n A_i) \leq \sum_i P(A_i)$  by induction from two-event inclusion-exclusion. When does equality hold? Give a computing use-case (e.g., bounding failure probability of  $n$  independent components).

## Chapter 3

# Conditional Probability & Bayes' Theorem

*“Information changes belief.”* You tested positive for a rare disease—how worried should you be? An email contains the word “lottery”—is it spam? Both are conditional probability questions:  $P(\text{truth} \mid \text{evidence})$ , not  $P(\text{evidence} \mid \text{truth})$ . Mixing the two is the most expensive mistake in applied statistics.

**From zero: we build on sample spaces (Ch. 2).** Conditional probability is just “zoom into the part of  $\Omega$  we now know occurred and renormalise.” Bayes’ rule flips the conditioning direction; independence means learning  $B$  tells you nothing about  $A$ .

## Learning Outcomes

After this chapter you will be able to:

- (L1) compute  $P(A \mid B)$  from the definition and interpret it as renormalisation;
- (L2) apply the law of total probability to partition evidence;
- (L3) state and apply Bayes’ theorem, including the odds form;
- (L4) recognise and test (pairwise/mutual) independence;
- (L5) diagnose base-rate fallacy and Simpson’s paradox in real data.

## 3.1 Conditional Probability

**Definition 3.1** (Conditional probability). For  $P(B) > 0$ ,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Read “probability of  $A$  given  $B$ ”.

*Picture:* condition on  $B$  = discard everything outside  $B$ , rescale what remains so  $B$  has total mass 1. Formally,  $P(\cdot | B)$  itself satisfies Kolmogorov's axioms on the new sample space  $B$ .

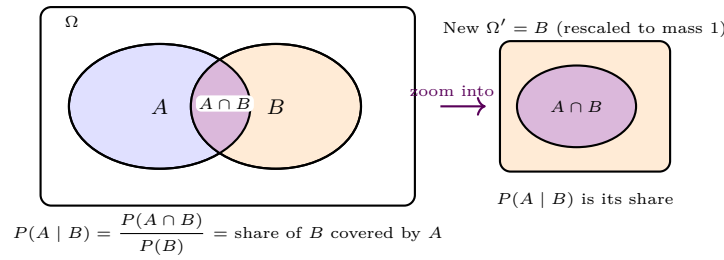


Figure 3.1: Conditioning on  $B$  zooms into  $B$  and renormalises.  $P(A | B)$  is the fraction of  $B$ 's mass that also lies in  $A$ .

**Example 3.1** (Two dice, given at least one 6).  $P(\text{sum } 7 | \text{at least one } 6) = |A \cap B|/|B| = 1/11$  (only  $(1, 6)$  qualifies among 11 outcomes with a 6). Unconditionally  $P(\text{sum } 7) = 1/6$ ; the information “a 6 appeared” changes the probability.

Properties:  $P(\cdot | B)$  is a probability measure; chain rule  $P(A \cap B) = P(A | B)P(B)$ ;  $P(A^c | B) = 1 - P(A | B)$ .

## 3.2 Law of Total Probability

If  $B_1, \dots, B_k$  partition  $\Omega$  (disjoint, union =  $\Omega$ , each  $P(B_i) > 0$ ),

$$P(A) = \sum_{i=1}^k P(A | B_i) P(B_i).$$

*Intuition:* slice  $\Omega$  into cases  $B_i$ , compute  $P(A)$  within each slice, weight by slice size. This is how we handle “ $A$  can happen via several mutually exclusive pathways.”

**Example 3.2** (Email spam via law of total probability). 60% of mail is spam ( $S$ ), 40% ham ( $H$ ). Word “lottery” appears in 30% of spam, 2% of ham. Then

$$P(\text{“lottery”}) = 0.30 \cdot 0.60 + 0.02 \cdot 0.40 = 0.188.$$

## 3.3 Bayes' Theorem

**Theorem 3.1** (Bayes' theorem). For  $P(B) > 0$ ,

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_i P(A | B_i) P(B_i)} = \frac{P(A | B_j) P(B_j)}{P(A)}.$$

It *inverts* conditioning: from  $P(\text{evidence} | \text{hypothesis})$  and prior  $P(\text{hypothesis})$  to posterior  $P(\text{hypothesis} | \text{evidence})$ .

**Corollary 3.1** (Odds form).  $\frac{P(B | A)}{P(B^c | A)} = \frac{P(A | B)}{P(A | B^c)} \cdot \frac{P(B)}{P(B^c)}$ . Posterior odds = likelihood ratio  $\times$  prior odds.



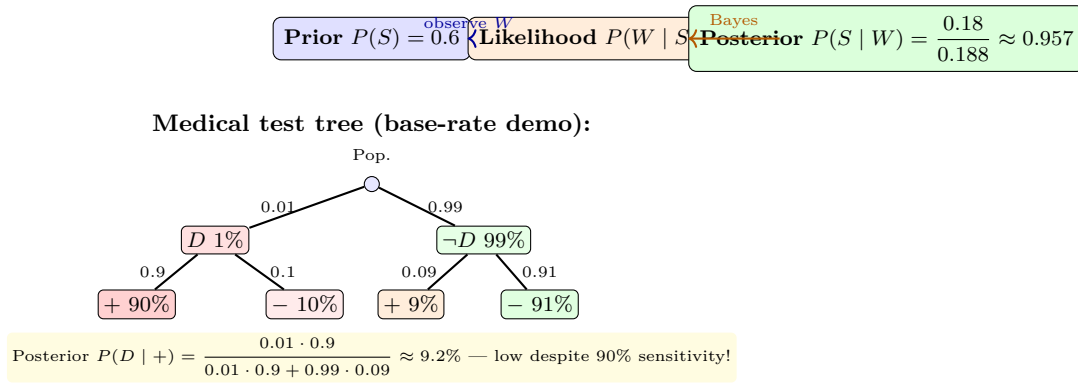


Figure 3.2: Top: Bayes as prior  $\rightarrow$  likelihood  $\rightarrow$  posterior. Bottom: medical test tree showing the base-rate fallacy.

**Example 3.3** (Rare disease, base-rate fallacy). Disease prevalence 1%, test sensitivity 90%, false-positive rate 9%. Then

$$P(D | +) = \frac{0.9 \cdot 0.01}{0.9 \cdot 0.01 + 0.09 \cdot 0.99} \approx 0.092.$$

Even after testing positive, only  $\approx 9\%$  chance of disease! The error is confusing  $P(+ | D) = 0.9$  with  $P(D | +)$ . Always account for the base rate  $P(D)$ .

### 3.4 Independence

**Definition 3.2** (Independence). Events  $A, B$  are *independent* if  $P(A \cap B) = P(A)P(B)$ , equivalently  $P(A | B) = P(A)$  when  $P(B) > 0$ . A family is *mutually independent* if every finite subcollection factorises; *pairwise independent* is weaker.

*Computing relevance:* independent failures multiply:  $P(\text{all } n \text{ replicas fail}) = p^n$  if each fails with prob.  $p$  independently. Hash collisions, randomised algorithms, and naive Bayes classifiers all assume (or approximate) independence.

**Example 3.4** (Pairwise but not mutual). Two fair coin tosses:  $A = \{\text{first } H\}$ ,  $B = \{\text{second } H\}$ ,  $C = \{\text{tosses agree}\}$ . Each pair independent ( $1/4 = 1/2 \cdot 1/2$ ), but  $P(A \cap B \cap C) = 1/4 \neq 1/8$ . Pairwise  $\nRightarrow$  mutual.

### 3.5 Simpson's Paradox (Preview)

Aggregated data can reverse trends seen in every subgroup. E.g., treatment  $A$  beats  $B$  in both men and women separately, yet  $B$  beats  $A$  overall—because group sizes differ (a conditioning effect). Always condition on relevant covariates; see Ch. 8 for regression's resolution.

**Example 3.5** (Simpson's numbers). Department X: treatment  $A$  cures 80/100 (80%) vs.  $B$  30/40 (75%); department Y:  $A$  20/40 (50%) vs.  $B$  10/100 (10%). In each department  $A$  wins, but overall  $A$  is 100/140  $\approx 71\%$  while  $B$  is 40/140  $\approx 29\%$  here  $A$  still wins—to flip it, swap group sizes: if  $A$  treated mostly hard cases, aggregate can reverse. Real Berkeley admissions (1973) showed exactly this structure.

### 3.6 Computing Bayes in Code

```

1 # Bayes: spam filter and medical test
2 def bayes_posterior(prior, likelihood_pos, false_pos):
3     """P(D|+) from prior P(D), P(+|D), P(+|~D)."""
4     num = likelihood_pos * prior
5     den = likelihood_pos * prior + false_pos * (1 - prior)
6     return num / den
7
8 print(f"P(spam|'lottery') = {bayes_posterior(0.6, 0.30, 0.02):.3f}") # 0.957
9 print(f"P(disease|+) = {bayes_posterior(0.01, 0.90, 0.09):.3f}")      # 0.092
10
11 # Chain rule demo: P(A,B,C) = P(A)*P(B|A)*P(C|A,B)
12 # Naive Bayes assumes P(features|class) factorises

```

### 3.7 Chapter Notes

Bayes' theorem appears in Bayes (1763, posthumous) and Laplace (1774). The base-rate fallacy is Kahneman–Tversky's classic; Gerd Gigerenzer advocates natural frequencies (“9 of 98 positives are diseased”) to make Bayes intuitive. Naive Bayes classifiers (Ch. 8) exploit conditional independence for text classification despite the “naive” assumption.

### 3.8 Exercises

- E3.1 **Conditional drill.** Two fair dice are rolled. Compute (a)  $P(\text{sum } 8 \mid \text{at least one } 4)$ , (b)  $P(\text{doubles} \mid \text{sum} \leq 6)$ . Draw the relevant subgrids.
- E3.2 **Law of total probability.** A factory has machines  $M_1, M_2, M_3$  producing 30%, 50%, 20% of chips with defect rates 2%, 1%, 5%. A random chip is defective—what is  $P(M_2 \mid \text{defective})$  via Bayes?
- E3.3 **Base-rate fallacy.** An intrusion detector has 99% detection rate and 1% false-alarm rate. Attacks occur in 0.1% of connections. Compute  $P(\text{attack} \mid \text{alarm})$ . How many alarms are false? Propose how to improve precision.
- E3.4 **Independence.** (a) Prove if  $A, B$  independent then  $A^c, B$  independent. (b) Construct three events pairwise independent but not mutually independent (verify all conditions numerically).
- E3.5 **Odds form.** Derive the odds form of Bayes. A spam filter has prior odds 3:2 for spam and likelihood ratio 15 for word “prize”. Compute posterior odds and convert to probability.

## Chapter 4

# Random Variables & Distributions

“A random variable is not random and not a variable.” It is a function  $X : \Omega \rightarrow \mathbb{R}$  that *measures* an outcome numerically. Once we have  $X$ , we can talk about its distribution, average, and spread—without ever mentioning  $\Omega$  again.

**From zero: outcomes  $\rightarrow$  numbers.** We define random variables, then build the four workhorses of computing—Bernoulli, Binomial, Poisson, Normal—each from intuition (coin, counts, rare events, bell curve) before formulas. Both discrete and continuous cases are covered.

### Learning Outcomes

After this chapter you will be able to:

- (L1) define random variables, PMFs/PDFs, and CDFs and translate between them;
- (L2) recognise Bernoulli, Binomial, Poisson, and Normal models and when each applies;
- (L3) compute probabilities via PMF/PDF and CDF for each family;
- (L4) use linearity-free facts (e.g., Poisson additivity, Normal standardisation);
- (L5) approximate Binomial by Poisson or Normal using the right regime.

### 4.1 Random Variables, PMF, PDF, CDF

**Definition 4.1** (Random variable). A *random variable* (r.v.)  $X$  is a function  $X : \Omega \rightarrow \mathbb{R}$ . For discrete  $X$ , the *probability mass function* (PMF) is  $p_X(k) = P(X = k)$ . For continuous  $X$ , the *probability density function* (PDF)  $f_X$  satisfies  $P(a \leq X \leq b) = \int_a^b f_X(x) dx$ . The *cumulative distribution function* (CDF) is  $F_X(t) = P(X \leq t)$  for either type.

PMF:  $p_X \geq 0$ ,  $\sum_k p_X(k) = 1$ . PDF:  $f_X \geq 0$ ,  $\int f_X = 1$ ; note  $P(X = t) = 0$  for continuous  $X$ , only intervals have mass. CDF is nondecreasing, right-continuous,  $F(-\infty) = 0$ ,  $F(+\infty) = 1$ .

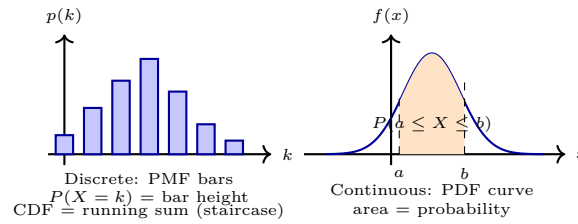


Figure 4.1: Left: discrete PMF (bar heights sum to 1). Right: continuous PDF (area under curve = 1; shaded area =  $P(a \leq X \leq b)$ ).

## 4.2 Bernoulli and Binomial

**Definition 4.2** (Bernoulli).  $X \sim \text{Bernoulli}(p)$  if  $P(X = 1) = p$ ,  $P(X = 0) = 1 - p$  for  $0 \leq p \leq 1$ . Think: one biased coin toss; 1 = success.

**Definition 4.3** (Binomial).  $X \sim \text{Binomial}(n, p)$  if

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, \dots, n.$$

Think: number of successes in  $n$  independent Bernoulli( $p$ ) trials.

*Derivation:* choose which  $k$  of  $n$  trials succeed ( $\binom{n}{k}$  ways), each such pattern has probability  $p^k (1 - p)^{n-k}$  by independence, sum over patterns (disjoint).

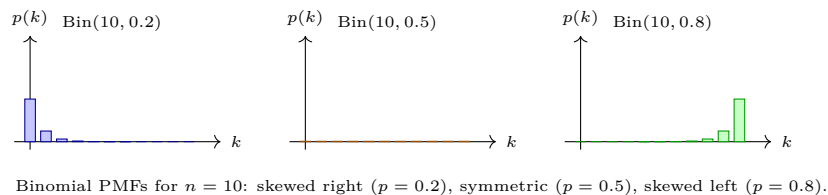


Figure 4.2: Binomial PMFs  $\text{Bin}(10, p)$  for three values of  $p$ . Shape moves from right-skewed to symmetric to left-skewed.

Mean and variance: if  $X \sim \text{Bin}(n, p)$ ,  $\mathbb{E}[X] = np$ ,  $\text{Var}(X) = np(1 - p)$  (proved in Ch. 5 via linearity and independence).

## 4.3 Poisson: Rare Events

**Definition 4.4** (Poisson).  $X \sim \text{Poisson}(\lambda)$  ( $\lambda > 0$ ) if

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Think: number of rare events in a large interval when events occur at rate  $\lambda$  independently.

*Why it appears:*  $\text{Bin}(n, \lambda/n) \rightarrow \text{Poisson}(\lambda)$  as  $n \rightarrow \infty$  (law of rare events). Use when  $n$  large,  $p$  small,  $np = \lambda$  moderate. Examples: packets arriving at a router per second, typos per page, clicks per minute.

Poisson is *additive*: if  $X \sim \text{Pois}(\lambda)$ ,  $Y \sim \text{Pois}(\mu)$  independent, then  $X + Y \sim \text{Pois}(\lambda + \mu)$ . Mean = variance =  $\lambda$ .

**Example 4.1** (Server requests). Requests arrive at 3/sec on average.  $P(\text{exactly 5 in a second}) = e^{-3}3^5/5! \approx 0.101$ .  $P(\geq 10)$  is the tail  $\sum_{k \geq 10} e^{-3}3^k/k! \approx 0.0011$ .

## 4.4 Normal (Gaussian): The Bell Curve

**Definition 4.5** (Normal).  $X \sim \mathcal{N}(\mu, \sigma^2)$  if PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

$\mu$  is the centre,  $\sigma$  the spread. Standard Normal  $Z \sim \mathcal{N}(0, 1)$ .

Key property: *standardisation*. If  $X \sim \mathcal{N}(\mu, \sigma^2)$  then  $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$ . So every Normal probability reduces to the standard CDF  $\Phi(z) = P(Z \leq z)$  (tabled or via **erf**).

68–95–99.7 rule:  $P(|Z| \leq 1) \approx 0.683$ ,  $P(|Z| \leq 2) \approx 0.954$ ,  $P(|Z| \leq 3) \approx 0.997$ .

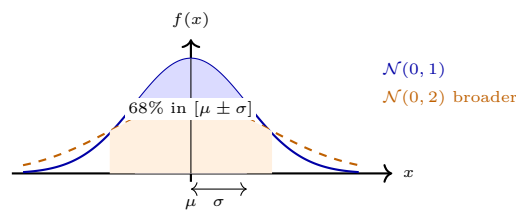


Figure 4.3: Normal PDFs: narrow ( $\sigma = 1$ ) vs. broad ( $\sigma \approx 1.45$ ). Shaded  $1\sigma$  bands contain  $\approx 68\%$  of mass. Standardisation maps any  $\mathcal{N}(\mu, \sigma^2)$  to  $\mathcal{N}(0, 1)$ .

**Approximations.** For  $n$  large:  $\text{Bin}(n, p) \approx \text{Pois}(np)$  when  $p$  small;  $\text{Bin}(n, p) \approx \mathcal{N}(np, np(1-p))$  when  $np(1-p) \gg 1$  (de Moivre–Laplace / CLT preview). Use continuity correction  $P(a \leq X \leq b) \approx \Phi((b + 0.5 - \mu)/\sigma) - \Phi((a - 0.5 - \mu)/\sigma)$ .

**Other key distributions (used later).** *Geometric*  $\text{Geom}(p)$ :  $P(X = k) = (1-p)^{k-1}p$  ( $k \geq 1$ ), trials until first success,  $\mathbb{E} = 1/p$ , memoryless  $P(X > m+n \mid X > m) = P(X > n)$ . *Exponential*  $\text{Exp}(\lambda)$ :  $f(x) = \lambda e^{-\lambda x}$  ( $x \geq 0$ ), continuous analogue of Geometric, also memoryless, models inter-arrival times. *Uniform*  $\text{Unif}[a, b]$ : constant density  $1/(b-a)$ . Each appears in Ch. 5–8 for waiting times and random algorithms.

## 4.5 Computing Distributions in Code

```

1 import math
2 from math import comb, exp, sqrt, pi, erf
3
4 def binom_pmf(n, p, k): return comb(n,k)*p**k*(1-p)**(n-k)
5 def poisson_pmf(lam, k): return exp(-lam)*lam**k/math.factorial(k)

```

```

6 def normal_pdf(x, mu=0, sigma=1):
7     return exp(-0.5*((x-mu)/sigma)**2)/sqrt(2*pi*sigma**2)
8 def Phi(z): return 0.5*(1+erf(z/sqrt(2))) # standard Normal CDF
9
10 # Examples
11 print(f"Bin(10,0.3) P(X=3) = {binom_pmf(10,0.3,3):.4f}")
12 print(f"Pois(3) P(X=5) = {poisson_pmf(3,5):.4f}")
13 print(f"N(0,1) P(|Z|<=1) = {Phi(1)-Phi(-1):.4f}") # 0.6827
14 # Normal approx to Bin(100,0.5): P(45<=X<=55)
15 mu, sigma = 50, 5
16 print(f"Normal approx = {Phi((55.5-mu)/sigma)-Phi((44.5-mu)/sigma):.4f}")

```

## 4.6 Chapter Notes

Bernoulli (1713) and Poisson (1837) distributions are named for their discoverers; Gauss (1809) and de Moivre (1733) shaped the Normal. Knuth TAOCP Vol. 2 covers discrete distributions for random algorithms. The Normal's ubiquity is explained by the Central Limit Theorem (Ch. 6).

## 4.7 Exercises

- E4.1 **Bernoulli to Binomial.** Show from first principles that if  $X_i \sim \text{Bernoulli}(p)$  i.i.d. then  $S_n = \sum_{i=1}^n X_i \sim \text{Bin}(n, p)$ . Derive  $P(S_n = k)$  by counting outcome strings.
- E4.2 **Poisson additivity.** Let  $X \sim \text{Pois}(2)$ ,  $Y \sim \text{Pois}(3)$  independent. Compute  $P(X + Y = 4)$  two ways: via convolution sum and via  $\text{Pois}(5)$ . Verify they match numerically.
- E4.3 **Normal standardisation.**  $X \sim \mathcal{N}(100, 15^2)$  models IQ. Compute  $P(X > 130)$  and  $P(85 \leq X \leq 115)$  via  $\Phi$ . What percentile is 130?
- E4.4 **Approximation regimes.** For  $n = 100$ ,  $p = 0.02$  ( $\lambda = 2$ ), compare  $\text{Bin}(100, 0.02)$  PMF at  $k = 0, 1, 2, 3$  with  $\text{Pois}(2)$ . When does the Normal approximation fail here and why?
- E4.5 **CDF practice.** For  $X \sim \text{Bin}(10, 0.5)$ , compute  $F(3) = P(X \leq 3)$  exactly and via the Normal approximation with continuity correction. What is the absolute error?

## Chapter 5

# Expectation, Variance & Concentration

*“Expectation is what happens on average; variance is how far average can mislead you.”* An algorithm’s *expected* runtime may be  $O(n \log n)$  while its worst case is  $O(n^2)$  (quicksort). Without variance and tail bounds, “on average” is a half-truth.

**From zero: weighted averages.** Expectation is the centre of mass of a distribution; variance is its moment of inertia. We derive linearity (which needs no independence) and build Markov, Chebyshev, and Chernoff bounds—the tools that turn averages into guarantees.

## Learning Outcomes

After this chapter you will be able to:

- (L1) compute  $\mathbb{E}[X]$  and  $\text{Var}(X)$  from PMF/PDF or via indicators;
- (L2) apply linearity of expectation (with dependence) and variance addition (with independence);
- (L3) use LOTUS, covariance, and correlation correctly;
- (L4) apply Markov, Chebyshev, and Chernoff/Hoeffding tail bounds;
- (L5) analyse expected runtime of randomised algorithms (e.g., quicksort, hashing).

### 5.1 Expectation: Centre of Mass

**Definition 5.1** (Expectation). For discrete  $X$ ,  $\mathbb{E}[X] = \sum_k k P(X = k)$ . For continuous  $X$  with PDF  $f$ ,  $\mathbb{E}[X] = \int x f(x) dx$ , when absolutely convergent. More generally  $\mathbb{E}[g(X)] = \sum g(k)p(k)$  or  $\int g(x)f(x) dx$  (LOTUS: Law Of The Unconscious Statistician).

*Intuition:* place mass  $p(k)$  at position  $k$  on a rod;  $\mathbb{E}[X]$  is the balance point.

**Theorem 5.1** (Linearity of expectation). For any  $X, Y$  and constants  $a, b$ ,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$

with *no independence assumption*. Hence  $\mathbb{E}[\sum_{i=1}^n X_i] = \sum_i \mathbb{E}[X_i]$ .

This is the most-used trick in randomised analysis: break a count into indicators.

**Example 5.1** (Indicators). Let  $X_i = \mathbf{1}_{\{\text{trial } i \text{ succeeds}\}} \sim \text{Bernoulli}(p)$ . Then  $S_n = \sum X_i \sim \text{Bin}(n, p)$  and  $\mathbb{E}[S_n] = \sum \mathbb{E}[X_i] = np$  without summing the Binomial PMF. Similarly, coupon collector, hashing collisions, and quicksort comparisons all yield to indicators.

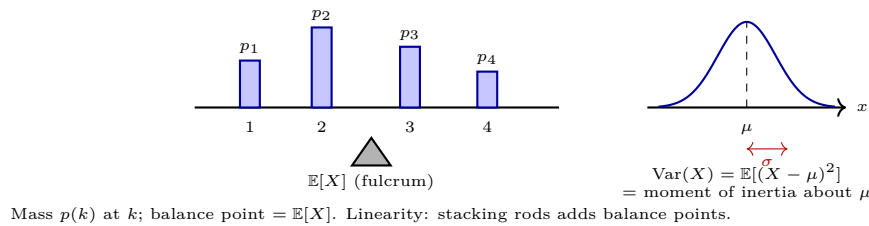


Figure 5.1: Left: expectation as centre of mass. Right: variance as spread (moment of inertia) about the mean.

Expectations of named distributions:  $\text{Bernoulli}(p) : p$ ,  $\text{Bin}(n, p) : np$ ,  $\text{Pois}(\lambda) : \lambda$ ,  $\mathcal{N}(\mu, \sigma^2) : \mu$ ,  $\text{Uniform}[a, b] : (a + b)/2$ ,  $\text{Geom}(p) : 1/p$ ,  $\text{Exp}(\lambda) : 1/\lambda$ .

*Remark 5.1* (Geometric vs. Exponential). Both are memoryless:  $P(X > m + n \mid X > m) = P(X > n)$ . Geometric counts discrete trials until success; Exponential measures continuous waiting time. They are the *only* memoryless laws in their classes—a fact used to justify Exponential inter-arrival models in queueing.

## 5.2 Variance, Covariance, Correlation

**Definition 5.2** (Variance).  $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 \geq 0$ . Standard deviation  $\sigma(X) = \sqrt{\text{Var}(X)}$ .

**Theorem 5.2** (Properties).  $\text{Var}(aX + b) = a^2 \text{Var}(X)$ . If  $X, Y$  independent,  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ . In general,

$$\text{Var}\left(\sum_i X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j),$$

where  $\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ . Correlation  $\rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y) \in [-1, 1]$ .

Independence  $\Rightarrow$  Cov = 0 (uncorrelated), but converse fails. Variance adds only under uncorrelatedness.

**Example 5.2** (Binomial variance).  $X = \sum_{i=1}^n X_i$  with  $X_i$  i.i.d.  $\text{Bernoulli}(p)$ , independent so  $\text{Var}(X) = np(1 - p)$ .

**Theorem 5.3** (Jensen's inequality). If  $g$  is convex,  $g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$ ; if concave, reverse. Example:  $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$  (since  $x \mapsto x^2$  convex), which is exactly  $\text{Var}(X) \geq 0$ . For concave log,  $\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$ —used in information theory (entropy bounds) and EM algorithms.



**Example 5.3** (Coupon collector expectation). Collect  $n$  coupons, each trial uniform over  $n$  types. Let  $T_k$  be time to get  $k$ th new coupon after  $k-1$  distinct seen;  $T_k \sim \text{Geom}(p_k)$  with  $p_k = (n-k+1)/n$ ,  $\mathbb{E}[T_k] = n/(n-k+1)$ . Total  $T = \sum_{k=1}^n T_k$ , so  $\mathbb{E}[T] = n \sum_{k=1}^n 1/k = nH_n \sim n \ln n$ . No joint distribution needed—linearity handles dependent stages via conditional expectations.

### 5.3 Tail Bounds: From Averages to Guarantees

**Theorem 5.4** (Markov). If  $X \geq 0$ , then for  $a > 0$ ,  $P(X \geq a) \leq \mathbb{E}[X]/a$ .

**Theorem 5.5** (Chebyshev). For any  $X$  with mean  $\mu$ , variance  $\sigma^2$ , and  $k > 0$ ,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}, \quad \text{or } P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

**Theorem 5.6** (Chernoff / Hoeffding, informal). For  $S_n \sim \text{Bin}(n, p)$  with  $\mu = np$  and  $0 < \delta < 1$ ,

$$P(S_n \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2 \mu}{3}\right), \quad P(|S_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2t^2}{n}\right) \text{ (Hoeffding)}.$$

Exponential tails vs. Chebyshev's  $1/t^2$ —vastly stronger for sums of bounded independent variables.

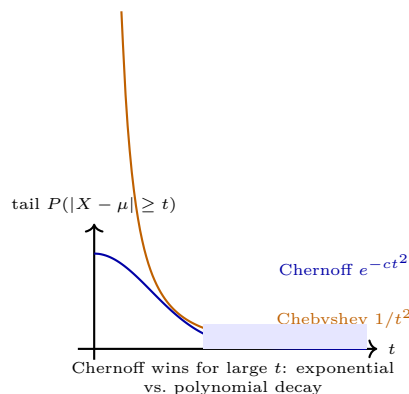


Figure 5.2: Chebyshev (polynomial) vs. Chernoff/Hoeffding (exponential) tail bounds. For sums of independent bounded variables, Chernoff dominates.

**Example 5.4** (Quicksort expected comparisons). Let  $X_{ij} = \mathbf{1}_{\{\text{elements } i, j \text{ compared}\}}$ . Then  $P(X_{ij} = 1) = 2/(j - i + 1)$  (they meet iff pivot is one of them first among  $[i..j]$ ). Total comparisons  $C = \sum_{i < j} X_{ij}$ , so

$$\mathbb{E}[C] = \sum_{i < j} \frac{2}{j - i + 1} = 2 \sum_{k=1}^{n-1} \frac{n - k}{k + 1} = 2(n + 1)H_n - 4n \sim 2n \ln n.$$

No independence needed—linearity suffices.

### 5.4 Correlation, Covariance Matrices, and the Bias–Variance Lens

For a random vector  $(X_1, \dots, X_d)$ , the *covariance matrix*  $\Sigma_{ij} = \text{Cov}(X_i, X_j)$  is symmetric positive semidefinite; diagonal entries are variances. Standardising gives the *correlation matrix*  $R_{ij} = \rho_{X_i, X_j}$ . In computing,  $\Sigma$

summarises feature redundancy: near-zero off-diagonals mean independent features, large  $|\rho|$  suggests one can be dropped or merged. PCA finds the orthogonal rotation diagonalising  $\Sigma$ , ordering directions by variance.

**Example 5.5** (Hash probe length). With linear probing, probe count  $P$  depends on load  $\alpha$ .  $\mathbb{E}[P] \approx \frac{1}{2}(1 + 1/(1 - \alpha))$  for unsuccessful search;  $\text{Var}(P)$  grows sharply as  $\alpha \rightarrow 1$ , so expectation alone understates tail latency—variance-aware sizing matters.

*Remark 5.2* (Why uncorrelated  $\neq$  independent matters for ML). Many algorithms whiten data (decorrelate) and assume independence follows; Ch. 6 shows it does not. Always test joint factorisation, not just  $\text{Cov} = 0$ , before assuming Naive Bayes or mean-field approximations.

## 5.5 Computing Expectation in Code

```

1  import random, math
2
3  # Monte Carlo: estimate E[X] and Var(X) for Bin(100,0.3)
4  n, p, trials = 100, 0.3, 200000
5  samples = [sum(random.random()<p for _ in range(n)) for _ in range(trials)]
6  mean = sum(samples)/trials
7  var = sum((x-mean)**2 for x in samples)/trials
8  print(f"Empirical mean {mean:.2f} (true {n*p}), var {var:.2f} (true {n*p*(1-p):.2f})")
9
10 # Chebyshev vs Chernoff for Bin(100,0.5), t=15
11 mu, sigma2 = 50, 25
12 t = 15
13 print(f"Chebyshev bound P(|X-mu|>=15) <= {sigma2/t**2:.4f}")
14 print(f"Hoeffding bound <= {2*math.exp(-2*t**2/n):.4f}") # 0.022
15 # Coupon collector: E[T] = n*H_n
16 import math as _m
17 for n in [10, 100, 1000]:
18     Hn = sum(1/k for k in range(1, n+1))
19     print(f"n={n} E[T]={n*Hn:.1f}")

```

## 5.6 Chapter Notes

Linearity of expectation is de Moivre’s insight; the indicator method is modern randomised-algorithms folklore (Motwani–Raghavan). Chebyshev (1867) and Chernoff (1952) bounds underpin PAC learning and randomised rounding. Quicksort analysis is Hoare (1962) refined by Sedgewick.

## 5.7 Exercises

**E5.1 Indicators.**  $n$  balls into  $n$  bins uniformly. Let  $X$  be the number of empty bins. Write  $X = \sum X_i$  with indicators, compute  $\mathbb{E}[X]$  and (harder)  $\text{Var}(X)$  via covariances. What happens as  $n \rightarrow \infty$ ?

**E5.2 LOTUS.**  $X \sim \text{Uniform}[0, 1]$ . Compute  $\mathbb{E}[X^2]$ ,  $\text{Var}(X)$ , and  $\mathbb{E}[e^X]$  via LOTUS. Verify  $\text{Var}(X) = 1/12$ .

- E5.3 **Markov vs. Chebyshev vs. Chernoff.** For  $S \sim \text{Bin}(100, 0.5)$ , bound  $P(S \geq 70)$  with each inequality. Compute the exact tail via code and compare tightness. Which bound is closest?
- E5.4 **Correlation trap.** Give  $X, Y$  dependent but  $\text{Cov}(X, Y) = 0$  (hint:  $X \sim \text{Uniform}\{-1, 0, 1\}$ ,  $Y = X^2$ ). Prove dependence yet zero correlation. What does this warn about “uncorrelated = independent”?
- E5.5 **Quicksort variance (challenge).** Using the same  $X_{ij}$  indicators, argue  $\text{Var}(C) = O(n^2)$  so Chebyshev gives concentration of  $C$  around  $2n \ln n$  within  $O(n)$  with high probability.

## Chapter 6

# Joint Distributions & the Central Limit Theorem

*“One variable tells a story; two variables tell the truth.”* Response time and server load, accuracy and training size, height and weight—computing data is multivariate. Joint distributions describe how variables move together, and the Central Limit Theorem explains why averages of many small effects are always Normal.

**From zero: pairs of outcomes.** We extend PMF/PDF to two variables, define joint, marginal, and conditional distributions, then build covariance through to the CLT—the most important theorem in statistics.

## Learning Outcomes

After this chapter you will be able to:

- (L1) write joint, marginal, and conditional PMFs/PDFs and translate between them;
- (L2) test independence via factorisation and compute Cov,  $\rho$ ;
- (L3) apply the weak and strong laws of large numbers;
- (L4) state and use the Central Limit Theorem (CLT) with continuity correction;
- (L5) recognise when CLT justifies Normal approximations for averages and sums.

## 6.1 Joint, Marginal, and Conditional Distributions

**Definition 6.1** (Joint PMF/PDF). For discrete  $(X, Y)$ ,  $p_{X,Y}(x, y) = P(X = x, Y = y)$ . For continuous,  $f_{X,Y}$  with  $P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$ . Marginals:  $p_X(x) = \sum_y p_{X,Y}(x, y)$  (or  $\int f_{X,Y} dy$ ). Conditional:  $p_{X|Y}(x | y) = p_{X,Y}(x, y)/p_Y(y)$ .

*Table intuition:* a joint PMF is a matrix; rows sum to  $p_Y$ , columns to  $p_X$ . Conditioning on  $Y = y$  picks one row and renormalises.

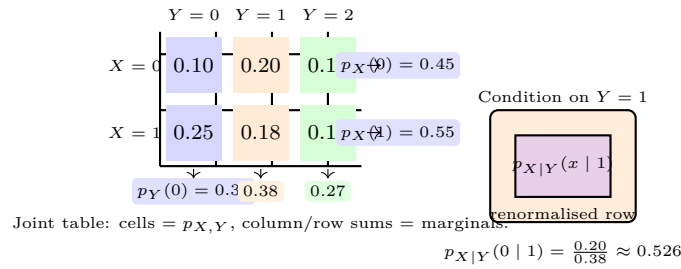


Figure 6.1: Joint PMF as a table: marginals are row/column sums; conditioning picks a row/column and renormalises.

**Definition 6.2** (Independence).  $X, Y$  independent iff  $p_{X,Y}(x, y) = p_X(x)p_Y(y)$  (or  $f_{X,Y} = f_X f_Y$ ) for all  $x, y$ . Equivalently  $F_{X,Y}(x, y) = F_X(x)F_Y(y)$ .

## 6.2 Covariance and Correlation Revisited

For jointly distributed  $(X, Y)$  with means  $\mu_X, \mu_Y$ ,

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X \mu_Y, \quad \rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

Cauchy–Schwarz gives  $|\text{Cov}| \leq \sigma_X \sigma_Y$ . If independent,  $\text{Cov} = 0$  (but not conversely—see Ch. 5 Ex. 4).

*Computing use:* covariance matrices summarise feature relationships; diagonal = variances, off-diagonal = covariances. PCA diagonalises this matrix.

## 6.3 Laws of Large Numbers

**Theorem 6.1** (Weak Law of Large Numbers, WLLN). If  $X_i$  i.i.d. with mean  $\mu$  and finite variance, sample mean  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$  satisfies for every  $\varepsilon > 0$ ,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \xrightarrow{n \rightarrow \infty} 0.$$

Proof via Chebyshev:  $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \text{Var}(\bar{X}_n)/\varepsilon^2 = \sigma^2/(n\varepsilon^2)$ .

**Theorem 6.2** (Strong Law, SLLN). Under same conditions,  $P(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$  (almost sure convergence).

*Meaning:* Monte Carlo works. Averaging many i.i.d. simulation runs converges to the true expectation. This justifies empirical estimation throughout computing.

## 6.4 Central Limit Theorem

**Theorem 6.3** (Central Limit Theorem, Lindeberg–Lévy). If  $X_i$  i.i.d. with mean  $\mu$  and variance  $\sigma^2 < \infty$ , then

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

That is, for large  $n$ ,  $\sum X_i \approx \mathcal{N}(n\mu, n\sigma^2)$  and  $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$ .

Why it matters: *any* sum/average of many small independent effects is approximately Normal, regardless of the underlying distribution (provided variance finite). Measurement errors, aggregated latencies, poll averages, stochastic gradient noise—all Normal by CLT.

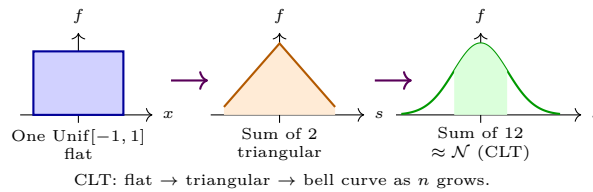


Figure 6.2: CLT in action: summing i.i.d. Uniform $[-1, 1]$  variables. One is flat, two is triangular, twelve is already bell-shaped.

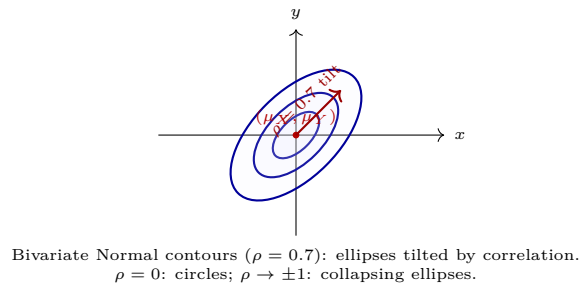


Figure 6.3: Bivariate Normal joint density contours. Correlation tilts and stretches the ellipses.

**Using CLT in practice.** To approximate  $P(a \leq S_n \leq b)$ : standardise  $Z = (S_n - n\mu)/(\sigma\sqrt{n})$ , then  $P \approx \Phi((b + 0.5 - n\mu)/(\sigma\sqrt{n})) - \Phi((a - 0.5 - n\mu)/(\sigma\sqrt{n}))$  with continuity correction for discrete  $S_n$ . Rule of thumb: CLT good when  $n \geq 30$  or  $np(1 - p) \geq 10$  for Binomial.

## 6.5 Computing Joint & CLT in Code

```

1 import random, math
2
3 # Sample mean and CLT demo: average of 12 Uniform -> approx Normal
4 def sample_mean(n, trials=80000):
5     return [sum(random.uniform(-1,1) for _ in range(n))/n for _ in range(trials)]
6
7 # WLLN: variance of mean shrinks as 1/n
8 for n in [1, 12, 100]:
9     means = sample_mean(n, 20000)
10    var = sum((x)**2 for x in means)/len(means) # mean is 0

```

```

11     print(f"n={n:3d}    Var(mean)={var:.5f}    theory={1/(3*n):.5f}")
12
13 # CLT: standardised sum approx N(0,1) P(|Z|<=1) ~ 0.6827
14 n = 50
15 sums = [sum(random.uniform(-1,1) for _ in range(n)) for _ in range(20000)]
16 sigma_sum = math.sqrt(n/3) # Var(Unif[-1,1])=1/3
17 inside = sum(abs(s)/sigma_sum <= 1 for s in sums)/len(sums)
18 print(f"CLT check P(|Z|<=1) empirical {inside:.3f} vs 0.683")

```

## 6.6 Chapter Notes

The WLLN is Bernoulli (1713) and Chebyshev (1867); SLLN is Kolmogorov (1933). De Moivre (1733) proved the first CLT (Binomial  $\rightarrow$  Normal); Lindeberg (1922) and Lévy gave the general form. The joint/marginal/conditional trilogy mirrors databases: joint = table, marginal = GROUP BY, conditional = WHERE + renormalise.

## 6.7 Exercises

- E6.1 **Joint table.** Discrete  $(X, Y)$  has  $p(0, 0) = 0.2$ ,  $p(0, 1) = 0.3$ ,  $p(1, 0) = 0.1$ ,  $p(1, 1) = 0.4$ . Compute marginals,  $p_{X|Y}(x | 1)$ ,  $\mathbb{E}[X]$ ,  $\mathbb{E}[Y]$ ,  $\text{Cov}(X, Y)$ ,  $\rho$ . Are  $X, Y$  independent?
- E6.2 **LLN vs. CLT.**  $X_i \sim \text{Bernoulli}(0.3)$  i.i.d. (a) Use WLLN to bound  $P(|\bar{X}_{100} - 0.3| \geq 0.1)$  via Chebyshev. (b) Use CLT to approximate the same probability. Which is tighter and why?
- E6.3 **Bivariate Normal.**  $(X, Y)$  bivariate Normal with  $\mu_X = \mu_Y = 0$ ,  $\sigma_X = \sigma_Y = 1$ ,  $\rho = 0.6$ . Write the joint PDF and describe the contour ellipses. What is the conditional distribution  $X | Y = y$ ?
- E6.4 **CLT approximation.**  $S_{100} \sim \text{Bin}(100, 0.5)$ . Approximate  $P(45 \leq S_{100} \leq 55)$  via CLT with continuity correction and compare to the exact sum via code.
- E6.5 **Simulation.** Simulate 100 000 sums of  $n = 30$   $\text{Exp}(1)$  variables (mean 1, var 1). Standardise and compare the empirical CDF at  $z = -1, 0, 1$  to  $\Phi(z)$ . How large must  $n$  be for good agreement when the underlying law is skewed?

# Chapter 7

## Estimation & Confidence Intervals

*“Data is not information until it is estimated with uncertainty.”* A single number (“average latency 42 ms”) is meaningless without “ $\pm 3$  ms with 95% confidence.” Estimation turns samples into guesses; confidence intervals quantify how much to trust them.

**From zero: samples  $\rightarrow$  guesses  $\rightarrow$  guarantees.** We assume i.i.d. samples from an unknown distribution. Point estimators guess the parameter; interval estimators add an error bar whose coverage probability we control. CLT (Ch. 6) powers the intervals.

### Learning Outcomes

After this chapter you will be able to:

- (L1) define bias, variance, and MSE and compute them for common estimators;
- (L2) derive MLEs for Bernoulli, Poisson, and Normal models;
- (L3) construct  $z$ - and  $t$ -confidence intervals for means and proportions;
- (L4) interpret “95% confidence” correctly (and spot misinterpretations);
- (L5) choose sample size to achieve a target margin of error.

### 7.1 Point Estimation: Bias, Variance, MSE

We observe i.i.d.  $X_1, \dots, X_n \sim f(x; \theta)$  with unknown  $\theta$ . An *estimator*  $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$  is a function of the data (hence random).

**Definition 7.1** (Bias, variance, MSE).  $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$ .  $\hat{\theta}$  is *unbiased* if  $\text{bias} = 0$ . Mean squared error

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + \text{Bias}^2.$$



Example: sample mean  $\bar{X} = \frac{1}{n} \sum X_i$  is unbiased for  $\mu$  ( $\mathbb{E}[\bar{X}] = \mu$ ,  $\text{Var}(\bar{X}) = \sigma^2/n$ ). Sample variance  $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$  is unbiased for  $\sigma^2$  (Bessel's correction); the MLE variant with  $1/n$  is biased low.

Trade-off: shrinking an unbiased estimator toward 0 introduces bias but can reduce MSE (bias–variance trade-off, central to ML).

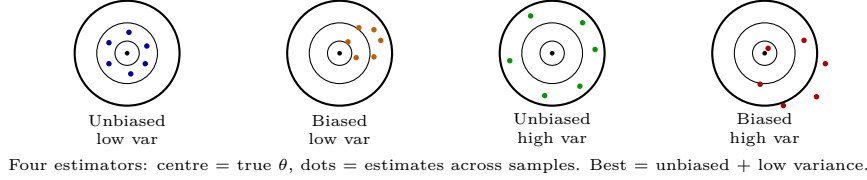


Figure 7.1: Bias–variance trade-off visualised as dartboards. Unbiased + low variance (leftmost) is ideal; real estimators trade one for the other.

## 7.2 Maximum Likelihood Estimation (MLE)

Likelihood  $L(\theta) = \prod_{i=1}^n f(X_i; \theta)$  (joint density as function of  $\theta$ ). Log-likelihood  $\ell(\theta) = \sum \log f(X_i; \theta)$  is easier to maximise.

**Definition 7.2** (MLE).  $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta)$ .

**Example 7.1** (Bernoulli MLE).  $X_i \sim \text{Bernoulli}(p)$ ,  $\ell(p) = S \log p + (n - S) \log(1 - p)$  where  $S = \sum X_i$ . Setting  $\ell'(p) = 0$  gives  $\hat{p} = S/n = \bar{X}$ , the sample proportion. Unbiased,  $\text{Var} = p(1 - p)/n$ .

**Example 7.2** (Normal MLE).  $X_i \sim \mathcal{N}(\mu, \sigma^2)$ :  $\hat{\mu} = \bar{X}$ ,  $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$  (biased; unbiased uses  $n - 1$ ).

**Example 7.3** (Poisson MLE).  $X_i \sim \text{Pois}(\lambda)$ :  $\hat{\lambda} = \bar{X}$ .

MLEs are consistent, asymptotically Normal, and asymptotically efficient (Cramér–Rao bound) under regularity.

**Example 7.4** (Method of moments vs. MLE). For  $X_i \sim \text{Unif}[0, \theta]$ , MoM gives  $\hat{\theta}_{\text{MoM}} = 2\bar{X}$  (since  $\mathbb{E}[X] = \theta/2$ ), while MLE is  $\hat{\theta}_{\text{MLE}} = \max_i X_i$  (likelihood  $1/\theta^n$  for  $\theta \geq \max X_i$ , 0 otherwise). MLE has smaller MSE here but is biased low ( $\mathbb{E}[\max] = \frac{n+1}{n+2}\theta$ ); bias-correcting gives  $\frac{n+1}{n} \max X_i$ , which is unbiased and dominates MoM in MSE.

## 7.3 Confidence Intervals

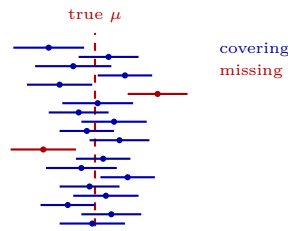
**Definition 7.3** (Confidence interval). A  $1 - \alpha$  confidence interval (CI)  $[L(X), U(X)]$  satisfies  $P_{\theta}(L \leq \theta \leq U) \geq 1 - \alpha$  for every  $\theta$ . After observing data,  $[l, u]$  either covers  $\theta$  or not—“95% confidence” refers to the *procedure*, not the realised interval.

### 7.3.1 CI for a Mean (Large $n$ or Known $\sigma$ )

By CLT,  $\bar{X} \approx \mathcal{N}(\mu, \sigma^2/n)$ . So an approximate  $1 - \alpha$  CI is

$$\bar{X} \pm z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}},$$

where  $z_{0.975} = 1.96$  for 95%. If  $\sigma$  unknown but  $n$  large, plug  $S$  for  $\sigma$  (Slutsky).



20 realised 95% CIs:  $\approx 19$  cover  $\mu$ ,  $\approx 1$  misses. “95%” = 95% of *procedures* cover, not 95% chance this one does.

Figure 7.2: Twenty 95% confidence intervals from independent samples. About 5% miss the true mean—exactly what 95% coverage means.

### 7.3.2 Small $n$ , Normal Data: $t$ -Interval

If  $X_i \sim \mathcal{N}(\mu, \sigma^2)$  with  $\sigma$  unknown and  $n$  small,

$$\bar{X} \pm t_{n-1, 1-\alpha/2} \cdot \frac{S}{\sqrt{n}},$$

with  $t_{n-1}$  (Student’s  $t$ , heavier tails than Normal). As  $n \rightarrow \infty$ ,  $t_{n-1} \rightarrow z$ .

### 7.3.3 CI for a Proportion

$\hat{p} = S/n$ ,  $\text{Var}(\hat{p}) = p(1-p)/n$ . Wald interval:  $\hat{p} \pm z_{1-\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$  (needs  $n\hat{p}(1-\hat{p}) \geq 5$ ). Better: Wilson or Clopper–Pearson for small  $n$ .

### 7.3.4 Sample-Size Planning

To get margin  $m = z \cdot \sigma / \sqrt{n} \leq m_0$ , need  $n \geq (z\sigma/m_0)^2$ . For proportions, worst-case  $p = 1/2$  gives  $n \geq (z/(2m_0))^2$ ; for  $m_0 = 0.03$  at 95%,  $n \geq 1068$ . For  $m_0 = 0.01$  at 95%,  $n \geq 9604$ —quadrupling precision quadruples  $n$  (CLT  $1/\sqrt{n}$  rate).

## 7.4 Bootstrap and Modern Intervals (Outlook)

When CLT conditions are doubtful (skewed data, small  $n$ , complex statistics), the *bootstrap* resamples the data with replacement to simulate the sampling distribution directly. Procedure: draw  $B$  bootstrap samples  $X_1^*, \dots, X_n^*$  from the empirical distribution, compute  $\hat{\theta}_b^*$  each time, then take quantiles of  $\{\hat{\theta}_b^*\}$ . No formula for  $\text{Var}(\hat{\theta})$  needed—the resamples estimate it. Bootstrap CIs (percentile, BCa) are standard in ML evaluation (e.g., confidence bands for accuracy). Cross-validation and permutation tests follow the same “simulate the null/sampling world” idea and connect to hypothesis testing (Ch. 8).

*Remark 7.1* (When to bootstrap vs.  $t$ ). Bootstrap wins for medians, trimmed means, or  $R^2$  where closed-form SE is messy;  $t/z$  win when Normality holds and  $n$  is moderate, giving shorter, exact intervals without resampling cost. Rule of thumb: if  $n \geq 30$  and data roughly symmetric,  $t$  is fine; otherwise bootstrap.

## 7.5 Computing Estimators in Code

```

1  import random, math
2
3  # Simulate CIs for Normal mean: coverage check
4  def ci_normal_mean(data, alpha=0.05):
5      n = len(data)
6      mean = sum(data)/n
7      s = math.sqrt(sum((x-mean)**2 for x in data)/(n-1))
8      z = 1.96 # for 95%, large n; use t_{n-1} for small n
9      half = z*s/math.sqrt(n)
10     return mean-half, mean+half
11
12 true_mu, true_sigma, n = 10.0, 2.0, 30
13 hits = 0
14 for _ in range(10000):
15     data = [random.gauss(true_mu, true_sigma) for _ in range(n)]
16     lo, hi = ci_normal_mean(data)
17     if lo <= true_mu <= hi: hits += 1
18 print(f"Empirical coverage: {hits/10000:.3f} (target 0.950)")
19
20 # MLE for Bernoulli: sample proportion
21 data_bern = [1,1,0,1,0,0,1,1,1,0]
22 p_hat = sum(data_bern)/len(data_bern)
23 print(f"Bernoulli MLE p_hat = {p_hat:.2f}")
24
25 # Bootstrap CI for the mean (percentile method)
26 def bootstrap_ci(data, B=5000, alpha=0.05):
27     n = len(data)
28     means = [sum(random.choices(data, k=n))/n for _ in range(B)]
29     means.sort()
30     lo = means[int((alpha/2)*B)]
31     hi = means[int((1-alpha/2)*B)]
32     return lo, hi
33 print("Bootstrap 95% CI:", bootstrap_ci([random.gauss(10,2) for _ in range(30)]))

```

## 7.6 Chapter Notes

Fisher (1922, 1925) created MLE and the modern CI framework (with Neyman, 1937). Student (Gosset, 1908) derived the  $t$ -distribution for small-sample brewing data. The “95% chance the true value is in this interval” misreading is the most common statistical error in computing papers—always say “95% of such intervals cover.”

## 7.7 Exercises

**E7.1 Bias and MSE.**  $X_i \sim \mathcal{N}(\mu, \sigma^2)$ . Compare estimators  $\hat{\mu}_1 = \bar{X}$  and  $\hat{\mu}_2 = \frac{1}{n+1} \sum X_i$  (shrunk). Compute bias, variance, MSE of each. For what  $\mu$  does  $\hat{\mu}_2$  win on MSE?

**E7.2 MLE for Poisson.** Derive the MLE  $\hat{\lambda}$  for i.i.d.  $\text{Pois}(\lambda)$  via log-likelihood. Is it unbiased? What is its

variance? Simulate  $n = 20$ ,  $\lambda = 3$  to verify.

**E7.3 CI construction.**  $n = 25$  Normal samples give  $\bar{x} = 42.3$ ,  $s = 5.1$ . Construct 95%  $t$ -CI for  $\mu$  (use  $t_{24,0.975} = 2.064$ ). How does it compare to the  $z$ -interval? Why is  $t$  wider?

**E7.4 Proportion CI.** In  $n = 200$  trials, 58 successes. Give 95% Wald CI for  $p$ . Check the  $n\hat{p}(1 - \hat{p})$  condition. What  $n$  is needed for margin  $\pm 0.02$  at 95% (worst case)?

**E7.5 Interpretation trap.** A 95% CI for latency is  $[41, 47]$  ms. A colleague says “ $P(41 \leq \mu \leq 47) = 0.95$ .” Explain why this is wrong, what is random vs. fixed, and how to phrase it correctly.

## Chapter 8

# Hypothesis Testing & Linear Regression

*“You cannot prove a hypothesis; you can only fail to reject it.”* Does the new cache policy really reduce miss rate? Is latency correlated with load? Hypothesis testing turns these questions into decisions with controlled error rates; regression quantifies the relationship.

**From zero: decisions under uncertainty.** We formalise null vs. alternative,  $p$ -values, Type I/II errors, then build  $z/t/\chi^2$  tests and simple linear regression from least squares—all with computing examples.

## Learning Outcomes

After this chapter you will be able to:

- (L1) formulate  $H_0$  vs.  $H_1$ , choose a test statistic, and compute  $p$ -values;
- (L2) control Type I/II errors, power, and significance level  $\alpha$ ;
- (L3) apply  $z$ -,  $t$ -, and  $\chi^2$  tests for means, proportions, and goodness-of-fit;
- (L4) derive and interpret simple linear regression (least squares,  $R^2$ );
- (L5) diagnose  $p$ -hacking, multiple testing, and practical vs. statistical significance.

## 8.1 The Logic of Hypothesis Testing

We have data  $X_1, \dots, X_n$  and two competing claims:

**Definition 8.1** (Hypotheses). *Null*  $H_0$  (status quo, e.g.,  $\mu = \mu_0$ ) vs. *alternative*  $H_1$  (what we seek evidence for:  $\mu \neq \mu_0$ ,  $\mu > \mu_0$ , or  $\mu < \mu_0$ ). A *test* is a rule: reject  $H_0$  if test statistic  $T(X)$  falls in a critical region.

**Definition 8.2** ( $p$ -value).  $p = P_{H_0}(T \text{ at least as extreme as observed})$ . Small  $p$  = data surprising under  $H_0$ . Reject  $H_0$  if  $p \leq \alpha$  (significance level, e.g., 0.05).

**Definition 8.3** (Errors). Type I: reject  $H_0$  when true (rate  $\alpha$ ). Type II: fail to reject when  $H_1$  true (rate  $\beta$ ).  $\text{Power} = 1 - \beta = P_{H_1}(\text{reject})$ . Increasing  $n$  or effect size raises power.

*Court analogy:*  $H_0$  = innocent,  $H_1$  = guilty. Type I = convict innocent; Type II = acquit guilty. “Not guilty”  $\neq$  “innocent”—failing to reject  $H_0$  does not prove it.

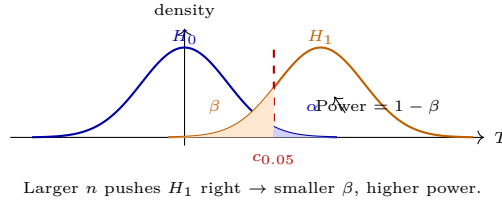


Figure 8.1: Null ( $H_0$ , blue) vs. alternative ( $H_1$ , orange) densities for test statistic  $T$ . Critical value  $c$  leaves  $\alpha$  in  $H_0$ 's tail;  $\beta$  is  $H_1$ 's mass before  $c$ ; power =  $1 - \beta$ .

## 8.2 Tests for Means and Proportions

**z-test (large  $n$  or known  $\sigma$ ).**  $Z = (\bar{X} - \mu_0)/(\sigma/\sqrt{n}) \approx \mathcal{N}(0, 1)$  under  $H_0$ . Two-sided  $p = 2(1 - \Phi(|z_{\text{obs}}|))$ .

**t-test (small  $n$ , Normal data, unknown  $\sigma$ ).**  $T = (\bar{X} - \mu_0)/(S/\sqrt{n}) \sim t_{n-1}$  under  $H_0$ . Use  $t$  critical values.

**Proportion test.**  $\hat{p} = S/n$ ,  $Z = (\hat{p} - p_0)/\sqrt{p_0(1 - p_0)/n} \approx \mathcal{N}(0, 1)$ .

**Two-sample t-test.** Compare means  $\mu_1, \mu_2$  via  $\bar{X}_1 - \bar{X}_2$  with pooled or Welch  $t$ .

**Example 8.1** (Cache policy). Old miss rate 12% ( $p_0 = 0.12$ ). New policy: 98 misses in 1000 accesses ( $\hat{p} = 0.098$ ).  $z = (0.098 - 0.12)/\sqrt{0.12 \cdot 0.88/1000} = -2.14$ ,  $p = 0.032 < 0.05$ : reject  $H_0$ , new policy significantly better (at 5%).

### 8.2.1 $\chi^2$ Goodness-of-Fit

Test if data follow a hypothesised distribution:  $\chi^2 = \sum_{i=1}^k (O_i - E_i)^2/E_i \approx \chi_{k-1}^2$  under  $H_0$ . E.g., is a die fair? Is packet size distribution Poisson?

## 8.3 Simple Linear Regression

We model  $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ ,  $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$  (or just mean 0, var  $\sigma^2$  for least squares).

**Definition 8.4** (Least squares). Choose  $\hat{\beta}_0, \hat{\beta}_1$  minimising  $\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$ .

Solution:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(Y_i - \bar{Y})}{\sum (x_i - \bar{x})^2} = \frac{\widehat{\text{Cov}}(x, Y)}{\widehat{\text{Var}}(x)}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}.$$

Fitted values  $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ , residuals  $e_i = Y_i - \hat{Y}_i$ .

**Definition 8.5** ( $R^2$ ).  $R^2 = 1 - \frac{\sum e_i^2}{\sum (Y_i - \bar{Y})^2} \in [0, 1]$ , fraction of variance explained.  $R^2 = r_{xY}^2$  (squared correlation) in simple regression.

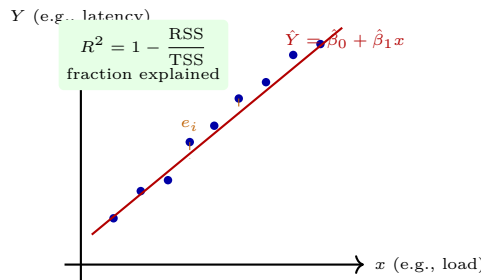


Figure 8.2: Simple linear regression: best-fit line minimises sum of squared residuals (dashed).  $R^2$  measures how much  $Y$  variation the line explains.

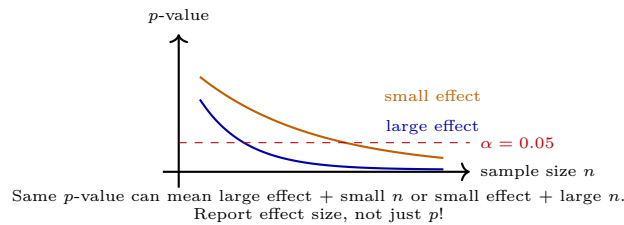


Figure 8.3:  $p$ -values conflate effect size and sample size. A tiny  $p$  with huge  $n$  may reflect a practically irrelevant effect—always report confidence intervals and effect sizes.

Inference: under Normal errors,  $\hat{\beta}_1 \sim \mathcal{N}(\beta_1, \sigma^2 / \sum (x_i - \bar{x})^2)$ , giving CI and  $t$ -test for  $H_0 : \beta_1 = 0$  (“is there a linear relationship?”).

## 8.4 Caveats: $p$ -Hacking and Practical Significance

*Multiple testing*: 20 tests at  $\alpha = 0.05$  give  $\approx 64\%$  chance of a false positive. Correct via Bonferroni ( $\alpha/k$ ) or FDR (Benjamini–Hochberg).  *$p$ -hacking*: trying many analyses until  $p < 0.05$  invalidates  $p$ . Pre-register hypotheses. *Practical vs. statistical*: with  $n = 10^6$ , a 0.01% improvement can be “significant” yet useless. Always report effect size and CI.

## 8.5 Computing Tests & Regression in Code

```

1 import math, random
2
3 def z_test_proportion(phat, p0, n):
4     se = math.sqrt(p0*(1-p0)/n)
5     z = (phat - p0)/se
6     # two-sided p-value via erf
7     Phi = lambda x: 0.5*(1+math.erf(x/math.sqrt(2)))
8     pval = 2*(1 - Phi(abs(z)))
9     return z, pval
10

```

```

11 z, pval = z_test_proportion(0.098, 0.12, 1000)
12 print(f"z={z:.2f} p={pval:.4f} {'reject H0' if pval<0.05 else 'fail to reject'}")
13
14 # Simple linear regression (closed form)
15 def linreg(x, y):
16     n = len(x)
17     mx, my = sum(x)/n, sum(y)/n
18     Sxx = sum((xi-mx)**2 for xi in x)
19     Sxy = sum((xi-mx)*(yi-my) for xi, yi in zip(x, y))
20     b1 = Sxy / Sxx
21     b0 = my - b1*mx
22     yhat = [b0 + b1*xi for xi in x]
23     RSS = sum((yi-yh)**2 for yi, yh in zip(y, yhat))
24     TSS = sum((yi-my)**2 for yi in y)
25     R2 = 1 - RSS/TSS
26     return b0, b1, R2
27
28 x = [1,2,3,4,5,6,7,8]
29 y = [2.1,3.9,6.2,7.8,10.1,11.9,14.2,16.0]
30 print("beta0,beta1,R2 =", linreg(x, y))

```

## 8.6 Chapter Notes

Neyman–Pearson (1933) formalised testing; Fisher advocated  $p$ -values. The replication crisis (Ioannidis 2005, Open Science Collaboration 2015) exposed  $p$ -hacking; modern guidance (ASA 2016, 2019) urges abandoning “ $p < 0.05$ ” as a bright line. Regression is Gauss (1809) and Legendre (1805); see Freedman *Statistical Models* for pitfalls and Simpson’s paradox (Ch. 3) as regression confounding.

## 8.7 Exercises

- E8.1  **$z$ -test.**  $n = 64$  packet delays give  $\bar{x} = 48.2$  ms,  $\sigma = 8$  ms known. Test  $H_0 : \mu = 50$  vs.  $H_1 : \mu \neq 50$  at  $\alpha = 0.05$ . Compute  $z$ ,  $p$ , decision, and 95% CI. Do CI and test agree?
- E8.2 **Power.** For  $H_0 : \mu = 0$  vs.  $H_1 : \mu = 0.5$  with  $\sigma = 1$ ,  $n = 25$ , one-sided  $\alpha = 0.05$ , compute power. How large must  $n$  be for power  $\geq 0.80$ ?
- E8.3  **$\chi^2$  fairness.** A die rolled 60 times gives counts [7, 12, 9, 14, 6, 12]. Test fairness at  $\alpha = 0.05$  ( $\chi^2_{5,0.95} = 11.07$ ). Compute  $\chi^2$  statistic and decide.
- E8.4 **Regression by hand.** Data  $(x, y)$ : (1, 2), (2, 3), (3, 5), (4, 4), (5, 6). Compute  $\hat{\beta}_0, \hat{\beta}_1, R^2$  via formulas. Predict at  $x = 6$  and give a residual plot interpretation.
- E8.5 **Multiple testing.** 100 independent true-null tests at  $\alpha = 0.05$ . What is  $P$ (at least one false positive)? What Bonferroni  $\alpha$  controls family-wise error at 0.05? Why is FDR less conservative?